proper device address and either receives the data being input or sends the data to be output.

Data can be input/output in either parallel or serial form. All data within a digital computer is represented in binary coded form. A binary data word consists of a group of bits; each bit is either a one or a zero. *Parallel* I/O consists of transferring all bits in the word at the same time, one bit per line. *Serial* I/O consists of transferring one bit at a time on a single line. Naturally serial I/O is much slower, but it requires considerably less hardware than does parallel I/O.

INTERRUPTS:

Interrupt provisions are included on many central processors, as a means of improving the processor's efficiency. Consider the case of a computer that is processing a large volume of data, portions of which are to be output to a printer. The CPU can output a byte of data within a single machine cycle but it may take the printer the equivalent of many machine cycles to actually print the character specified by the data byte. The CPU will have to remain idle waiting until the printer can accept the next data byte. If an interrupt capability is implemented on the computer, the CPU can output a data byte then return to data processing. When the printer is ready to accept the next data byte, it can request an interrupt. When the CPU acknowledges the interrupt, it suspends main program execution and automatically branches to a routine that will output the next data byte. After the byte is output, the CPU continues with main program execution. Note that this is, in principle, quite similar to a subroutine call, except that the jump is initiated externally rather than by the program.

More complex interrupt structures are possible, in which several interrupting devices share the same processor. Interruptive processing is an important feature that enables maximum utilization of a processor's capacity.

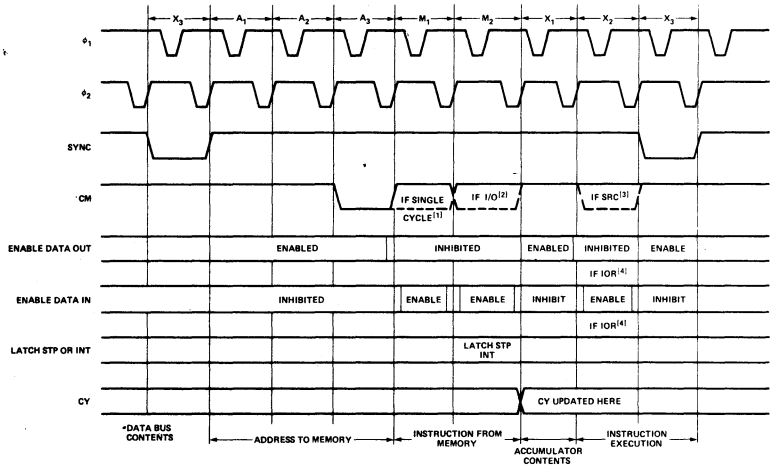
MCS-40™ SYSTEM OPERATION

The MCS-40 microcomputer, like all computer systems, contains a processor and memory. The processor (4040,4004) is contained in one dual in-line I.C. package. The memory can be either ROM program memory ro RAM data memory. In addition to information exchanges between the processor and memory, the processor also provides input and output (I/O) to external devices. The I/O faculties of the MCS-40 system are found in a separate I/O device or shared with memory elements, allowing for a reduction in package count. All MCS-40 inter-element communication transpires over a four bit data bus (D_0-D_3), D_3 being the most significant bus bit.

The data bus transfers information from the processor to the memory and I/O elements such as instruction addresses, operand addresses, operands, and I/O data. The processor receives instructions, operands, and I/O data back from the other elements. All traffic on the bus is contained within one instruction cycle for one cycle instructions. The instruction cycle is subdivided into equal time segments. Each segment is equivalent to one system clock period. Information on the data bus is altered from segment to segment. The first three time segments present a twelve bit (three groups of four bits) instruction address to the memory, least significant nibble first. The fourth and fifth segments provide the 8 bit instruction sequentially placed on the data bus by program memory. The sixth segment is utilized for instruction decode. The remaining two segments are used for program execution. Operands and I/O data can be found on the data bus during this time, depending on the instruction being executed.

Instructions requiring two cycles proceed in a similar manner as described above. Complete execution requires 16 clock periods which are partitioned into two eight segment cycles. During the first cycle the instruction will be fetched. Information fetched during this first cycle may also include a portion of the operand or an indirect address register. The second cycle will always fetch the operand and perform the execution.

Differentiation between one and two cycle instructions are performed by all elements on the data bus by decoding the unique instructions.



4040 Basic Timing Diagram